PART B: INDIVIDUAL REFLECTION PRESENTATION

Artificial Intelligence (AI) in the Enterprise

Innovation, Risk, and Responsible Governance

STUDENT ID
12703213

MODULE / COURSE
Launch Into Computing

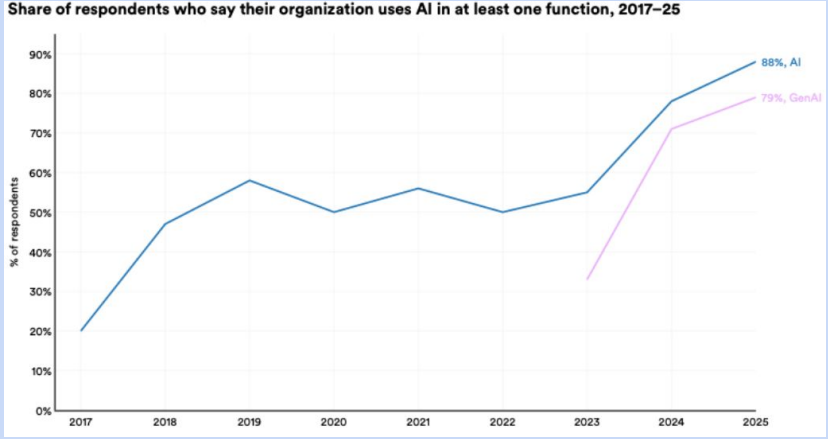
SUBMISSION DATE
July 2026

01 | Introduction: From Speculation to Reality

Scale of Adoption
AI adoption is accelerating, with organisational use rising from 20% to 88% between 2017 and 2025.

Enterprise Value
AI tools drive efficiency through automation, predictive analytics and personalisation.

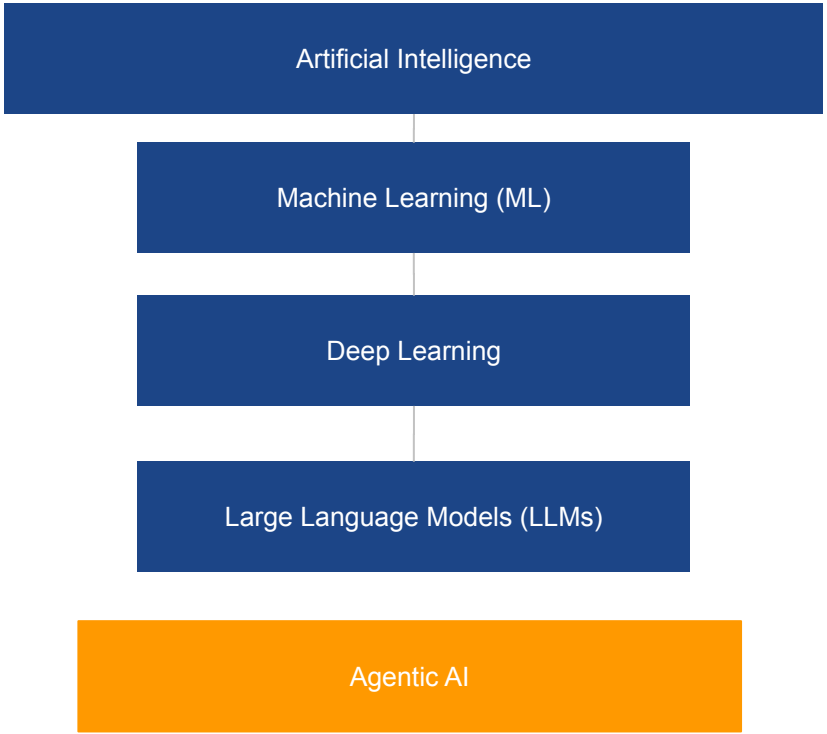
The Governance Gap
Governance frameworks and evaluation methods are struggling to keep pace with AI development, creating significant ethical, security, and accountability risks.



Source: (Stanford HAI, 2026, p. 193)

References: (Kneusel, 2023; Stanford HAI, 2026; Taeihagh, 2021)

02 | Technical Overview: What is AI?



Machine Learning
Learns patterns from data using supervised, unsupervised, reinforcement, and hybrid methods. E.g. fraud detection, recommendation engines, credit scoring.

LLMs
Models trained on large text-based datasets, enabling natural language understanding, generation, and reasoning. E.g. OpenAI's ChatGPT.

Agentic AI
Autonomous AI systems that orchestrate, act and coordinate tasks across workflows with varying levels of human oversight. E.g. LangGraph.

References: (Russell and Norvig, 2022)

03 | Enterprise Applications of AI

Opportunity and Governance Responsibility

The same AI applications that create business value can also introduce significant governance risks.

Area	AI Application	Business Value	Risk Introduced
Financial Services	Fraud Detection, Credit Risk Assessment, Financial Forecasting	Reduced fraud, faster and more consistent credit decisions	Bias; unfair and opaque modelling / Black-Box
Human Resources	CV Screening, Onboarding (e.g. Amazon)	Faster hiring and onboarding processes	Discrimination
Knowledge Management	Document Retrieval, Summarisation, Transcription	Faster access to internal knowledge	Data leakage; misinformation / hallucination

References: (Dastin, 2018; Mehrabi *et al.*, 2021)

04 | Ethical Issues

Ethical Principles Must Become Operational Controls

Using financial services as the example, the table below shows how core AI ethics principles translate into enterprise risks. The key challenge is turning ethical principles into controls that can be tested, audited, and governed.

Principle	Issue	Details	Consequence
Fairness	Algorithmic bias	Historical data can reproduce past discrimination	Unfair outcomes for protected groups
Accountability	Proxy discrimination	Neutral variables may indirectly encode protected traits	Bias can persist after protected data is removed
Explainability	Model opacity	Opaque models limit decision transparency	Decisions are harder to justify and challenge
Privacy	Data use and transparency	Personal data may be used without clear safeguards	Data-protection risk and reduced customer trust

References: (Leslie, 2019; Mehrabi *et al.*, 2021; Langenbucher and Corcoran, 2022; Bücker *et al.*, 2022; Floridi *et al.*, 2018)

05 | Security Concerns

When Safety Principles Are Not Operationally Enforced

Safe, explainable and accountable AI systems are essential for responsible deployment.

Concern	Threat	Impact	Operational Control
Adversarial Attacks	Manipulated inputs cause incorrect model behaviour	Fraud or credit-risk models may be bypassed	Adversarial testing and stress testing
Data Leakage	Inference attacks expose sensitive model data	Customer exposure and regulatory breach	Data minimisation, redaction and access controls
Hallucination Risk	LLMs generate plausible but unsupported content	False claims or citations in professional outputs	Source verification and human review
Model Poisoning	Corrupted training data distorts model behaviour	Unreliable outputs in high-impact decisions	Data validation and continuous monitoring

References: (Goodfellow, Shlens and Szegedy, 2015; Ji *et al.*, 2023)

06 | Regulation and Governance

Navigating Fragmented AI Governance

No single framework provides comprehensive AI governance, leaving enterprises to navigate a fragmented and changing regulatory landscape.

Framework	Purpose	Limitation	Enterprise Response
EU AI Act (2024)	Risk-based classification and obligations	EU-focused; phased implementation	Classify use cases and apply proportionate controls
GDPR / UK GDPR	Protects personal data and certain significant automated decisions	Not a complete AI governance framework	Ensure lawful processing, transparency and human intervention
UK FCA Consumer Duty (2023)	Requires good customer outcomes	Financial-services-specific	Test fairness, explainability and customer understanding
UK AI Assurance Guidance	Provides practical assurance techniques	Non-binding guidance	Use impact assessments, audits, red-teaming and monitoring

References: (Regulation (EU) 2024/1689, 2024, chapter III; Financial Conduct Authority, 2024; Department for Science, Innovation and Technology, 2023)

07 | Case Study: Deloitte Australia Gen-AI “Hallucination”

Background
A Deloitte assurance report for Australia’s Department of Employment and Workplace Relations was re-published after fabricated references and a false legal quotation were identified.

Consequence
- Reputational damage
- Partial refund / repayment of contract amounts
- Rework; report was re-published acknowledging the initial error

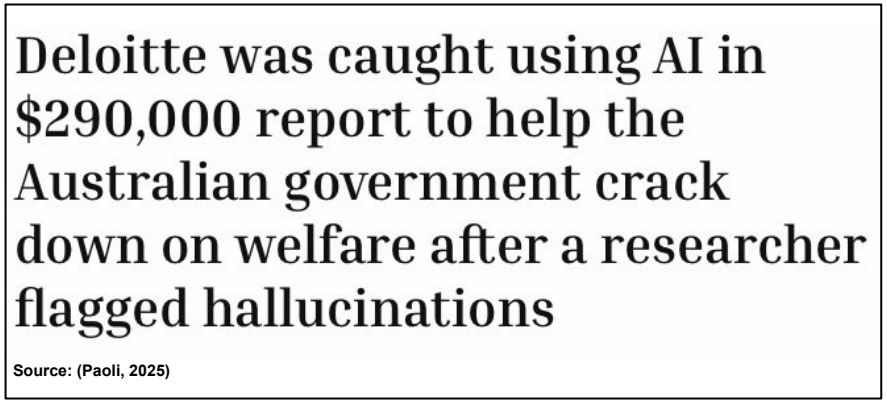
Implications for Practice
- Verify all AI-assisted outputs before publication
- Maintain audit trails and source traceability
- Disclose Gen-AI use in professional work and retain human accountability for final assurance

Key Lesson
Weak governance and assurance allowed AI-related risks to materialise at scale.


Deloitte issues refund for error-ridden Australian government report that used AI

Big Four firm will repay final instalment after incorrect references and citations found in document

Source: (Kissin, 2025)


Deloitte was caught using AI in $290,000 report to help the Australian government crack down on welfare after a researcher flagged hallucinations

Source: (Paoli, 2025)

References: (Deloitte Australia, 2025, p.147; Kissin, 2025; Paoli, 2025)

08 | Generative AI (Gen-AI) Response Summary

Prompt: *"... Critically evaluate one of these technologies and discuss how enterprises can balance innovation with security and ethical considerations..."*
Tool: Perplexity AI | **Generated**: 10 July 2026

Response format

- Quick content generation
- 4-part structure:
  - Introduction
  - Critical evaluation
  - Enterprise balance
  - Bottom line

Themes Identified

- Benefits: Improved Productivity, Decision-Making, and Threat Detection
- Innovation-governance trade-off
- Risks: bias, privacy and transparency
- Security: phishing, impersonation, data leakage and model poisoning
- Controls: secure-by-design, monitoring and human oversight

Note: AI-generated outputs used for comparison only.
See Appendices 1-3 for prompt, output and cited sources.

References: (Perplexity AI, 2026)

09 | Critical Comparison: Human Research vs. Perplexity AI (Gen-AI) Response

The Perplexity Gen-AI response identified relevant themes quickly but lacked academic depth, regulatory nuance and case study-based examples needed for professional judgement.

Area	Human Research	Perplexity AI (Gen-AI) Response
Source Reliability	Academic, regulatory and case-study sources	Mixed web/industry sources
Depth of Analysis	Risks linked to governance and practice	Broad summary; uniform structure appears template-led
Ethics and governance	Named ethical framework and regulatory analysis	General controls but no named framework or law
Case Study / Examples	Deloitte Australia used as a specific case study	No named case studies
Professional value	Verifiable and contextual but slower	Fast initial scoping but requires verification

Note: AI-generated outputs used for comparison only. See Appendices 1-3 for prompt, output and cited sources.

References: (Perplexity AI, 2026; Bender *et al*., 2021; Leslie, 2019)

10 | Conclusions and Recommendations

Conclusions

- AI improves productivity, decision-making and threat detection, but also scales risk
- Ethical risks such as bias, opacity, privacy loss and weak accountability are structural, not incidental
- The Deloitte Australia case shows AI failures can also be governance failures, not solely technical failures
- Gen-AI tools are useful for scoping but require human verification and professional judgement

Recommendations

- Treat AI governance as a design requirement, not an afterthought
- Apply risk-based controls to high-impact use cases
- Require bias testing, privacy review, red-teaming and audit trails
- Keep humans accountable for high-stakes decisions and published outputs

11 | References

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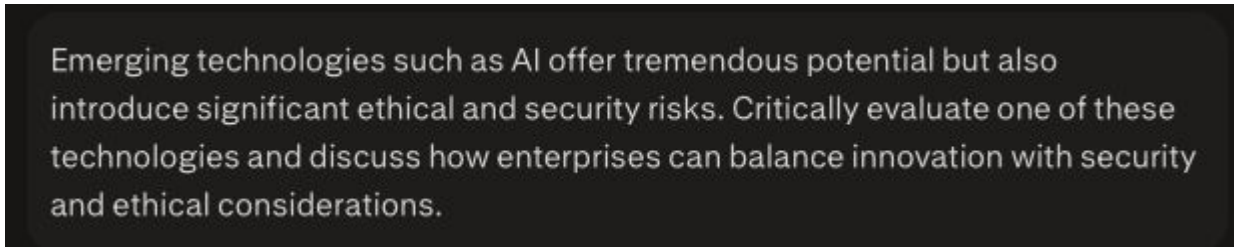
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Appendix 1 - Gen AI prompt Submitted

Prompt submitted to Perplexity AI.

Link: https://www.perplexity.ai/search/emerging-technologies-such-as-Ou.DJT9JQem8uDHCRMa7hg

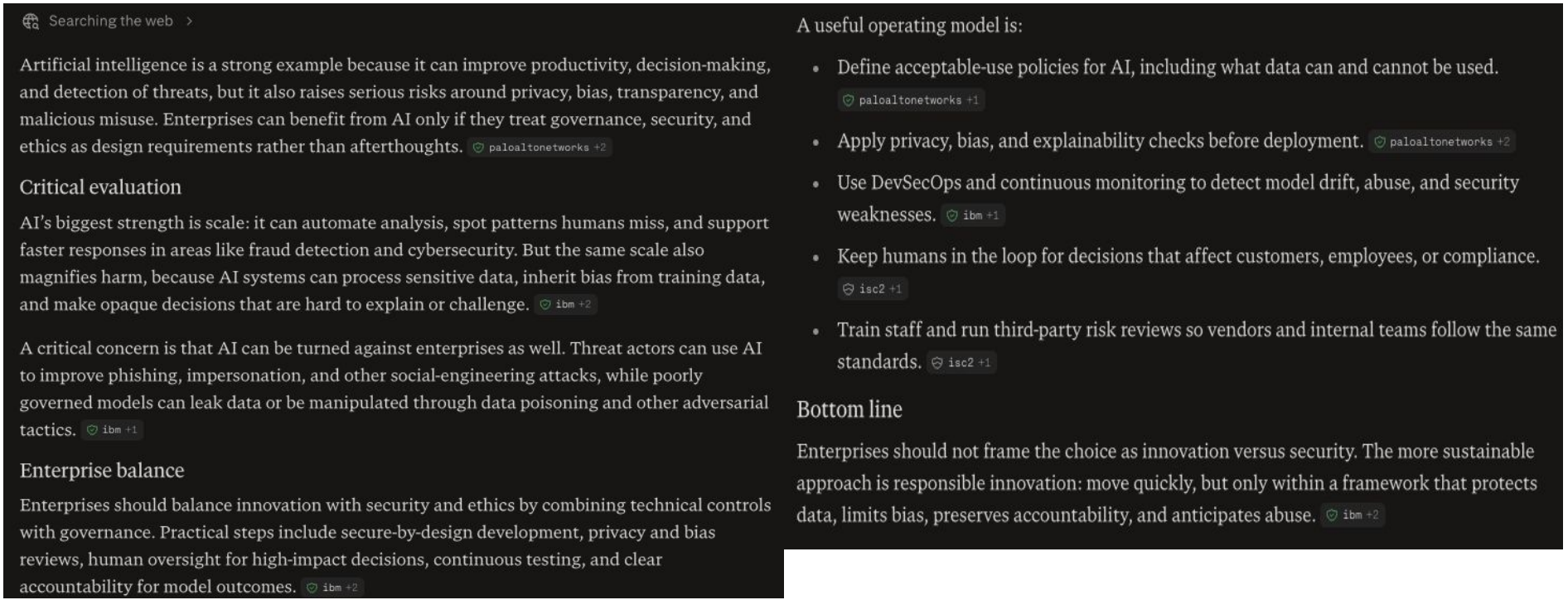


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Appendix 2 - Perplexity AI Response

Perplexity AI (2026) Perplexity response to Student 12703213. 10 July.

Link: https://www.perplexity.ai/search/emerging-technologies-such-as-Ou.DJT9JQem8uDHCRMa7hg



Searching the web

Artificial intelligence is a strong example because it can improve productivity, decision-making, and detection of threats, but it also raises serious risks around privacy, bias, transparency, and malicious misuse. Enterprises can benefit from AI only if they treat governance, security, and ethics as design requirements rather than afterthoughts. paloaltonetworks +2

Critical evaluation

AI's biggest strength is scale: it can automate analysis, spot patterns humans miss, and support faster responses in areas like fraud detection and cybersecurity. But the same scale also magnifies harm, because AI systems can process sensitive data, inherit bias from training data, and make opaque decisions that are hard to explain or challenge. ibm +2

A critical concern is that AI can be turned against enterprises as well. Threat actors can use AI to improve phishing, impersonation, and other social-engineering attacks, while poorly governed models can leak data or be manipulated through data poisoning and other adversarial tactics. ibm +1

Enterprise balance

Enterprises should balance innovation with security and ethics by combining technical controls with governance. Practical steps include secure-by-design development, privacy and bias reviews, human oversight for high-impact decisions, continuous testing, and clear accountability for model outcomes. ibm +2

A useful operating model is:

- Define acceptable-use policies for AI, including what data can and cannot be used. paloaltonetworks +1
- Apply privacy, bias, and explainability checks before deployment. paloaltonetworks +2
- Use DevSecOps and continuous monitoring to detect model drift, abuse, and security weaknesses. ibm +1
- Keep humans in the loop for decisions that affect customers, employees, or compliance. isc2 +1
- Train staff and run third-party risk reviews so vendors and internal teams follow the same standards. isc2 +1

Bottom line

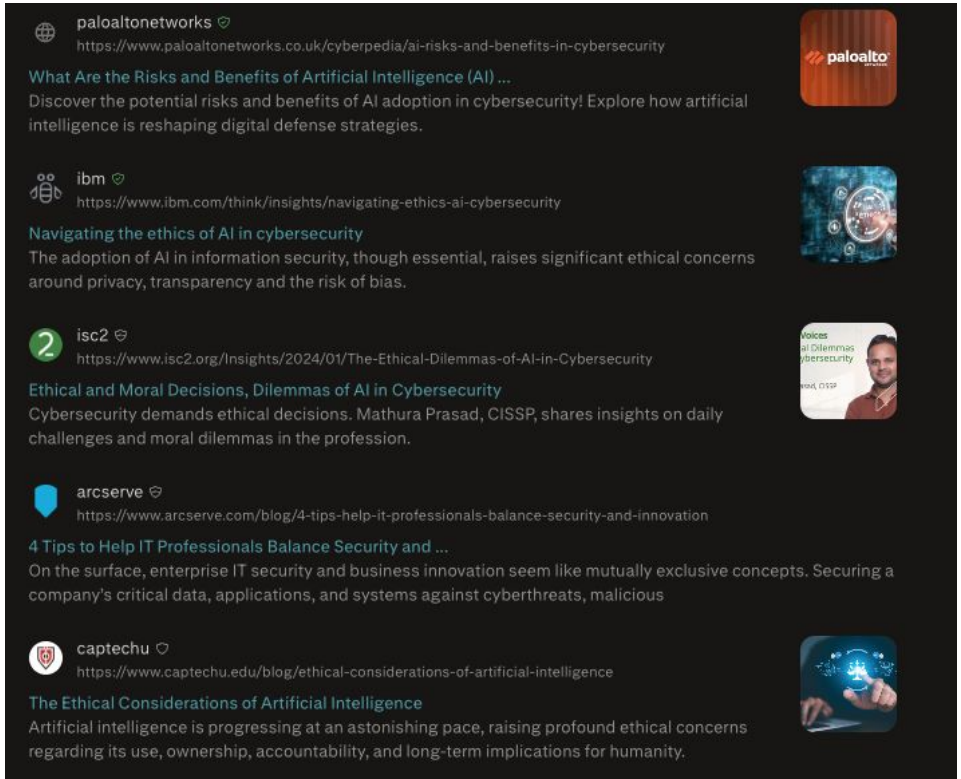
Enterprises should not frame the choice as innovation versus security. The more sustainable approach is responsible innovation: move quickly, but only within a framework that protects data, limits bias, preserves accountability, and anticipates abuse. ibm +2

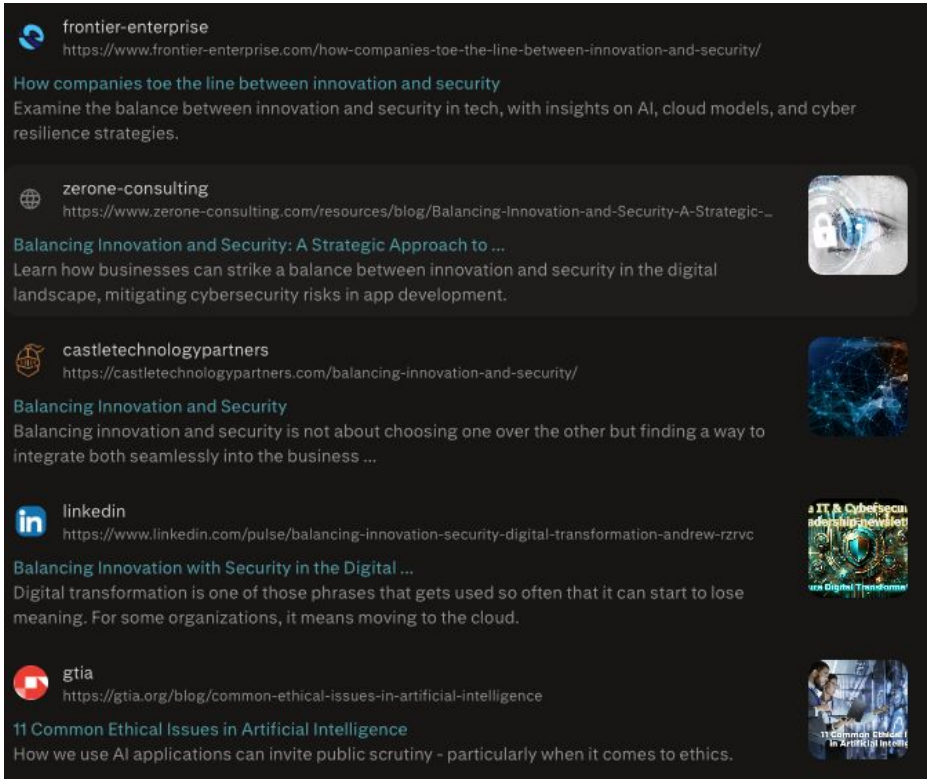
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Appendix 3 - Sources cited by Perplexity AI

Perplexity AI (2026) Perplexity response to Student 12703213. 10 July.

Link: https://www.perplexity.ai/search/emerging-technologies-such-as-Ou.DJT9JQem8uDHCRMa7hg





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Appendix 4 - Extended Enterprise Applications of AI

The below table extends the list of enterprise applications of AI discussed in Slide 03.

Area	AI Application	Business Value	Risk Introduced
Customer Service	Personalised Product Recommendations, Chatbots	Customer Personalisation, faster responses, 24/7 support	Hallucination; privacy concerns
Finance & Risk Management	Fraud Detection, Credit Risk Assessment, Financial Forecasting	Reduced fraud, faster and more consistent credit decisions	Bias; unfair and opaque modelling / Black-Box
Human Resources	CV Screening, Onboarding (e.g. Amazon)	Faster hiring and onboarding processes	Discrimination
Marketing & Sales	Recommendation Platforms, Campaign Generation (e.g. Netflix)	Better targeted campaigns	Privacy concerns
Cybersecurity	Threat detection, Incident Response	Faster detection	False negatives
Software Development	Code generation, code debugging, testing	Increased productivity	Insecure and unethical code
Knowledge Management	Document Retrieval, Summarisation, Transcription	Faster access to internal knowledge	Data leakage; misinformation / hallucination